EduMentor AI

Testing & Validation — Review Suggestions & Gap Analysis

Project	EduMentor AI
Stack	React + FastAPI + PostgreSQL + Google Gemini AI
Team	24UCS07, 24UCS13
Based on	EduMentor_Testing_Validation_Review.pdf (46 existing TCs)
Date	08-09-2026
Purpose	Identify missing tests & suggest improvements for Review 3

Existing TCs	Missing Tests Found	Improvement Areas	Recommended Total
46	27	5	81

A. What the Existing Document Does Well

The current Testing & Validation Review is well-structured and covers the core testing layers. The following strengths were identified:

Strength	Detail
4-level testing structure	Unit → Integration → Validation → System testing — proper layering
Equivalence partitioning	Year, Score, Email, Password, Topic fields all covered
Security tests included	SQL injection, XSS, role escalation, bcrypt hash check
Bug tracking section	5 bugs logged with location and fix status
Performance baselines	Measured actual vs target (login ~180ms, chat ~1.2s, dashboard ~950ms)
Validation against FRs	All 13 functional requirements traced to test cases

B. Missing Tests Found (Cross-checked with Source Code)

These features exist in the actual codebase but have zero test coverage in the current document.

B.1 — OTP / Forgot Password Flow (auth.py Lines 207–283)

■ CRITICAL: A full OTP-based forgot/reset password system exists in auth.py — 7 test cases are completely missing from the document.

TC#	What to Test	Input	Expected Output
UT-NEW-01	Forgot password — valid student email	POST /forgot-password + registered email	OTP sent, success: true
UT-NEW-02	Forgot password — unregistered email	Unknown email + role: student	404 Not Found
UT-NEW-03	Reset password — valid OTP	Correct OTP + new password (≥6 chars)	Password updated, OTP deleted from store
UT-NEW-04	Reset password — expired OTP	OTP used after 10-minute window	400 OTP has expired
UT-NEW-05	Reset password — wrong OTP	Incorrect 6-digit OTP	400 Invalid OTP
UT-NEW-06	Reset — password too short	new_password < 6 characters	400 Password must be at least 6 characters
UT-NEW-07	Forgot password — invalid role	role: admin (not student/mentor)	400 Invalid role

Example — UT-NEW-03 test verification:

```
POST /api/auth/reset-password
Body: { "email": "arun@edumentor.edu", "otp": "123456",
"new_password": "newpass123", "role": "student" }

Expected Response: { "success": true,
"message": "Password reset successful! You can now login..." }
Assert: student.password_hash in DB starts with '$2b$' (bcrypt)
```

B.2 — AI Quiz Generation Endpoint (students.py)

TC#	What to Test	Input	Expected Output
UT-NEW-08	Generate AI quiz for a topic	POST /students/generate-quiz + topic + subject (authenticated)	Returns MCQ, TF, fill-in, short questions
UT-NEW-09	Quiz generation — unauthenticated	No auth token	401 Unauthorized
UT-NEW-10	Quiz generation — empty topic	Blank topic field	422 Validation Error

B.3 — Quiz Status / Duplicate Submission Prevention (students.py)

Endpoint GET /api/students/quiz-status prevents students from submitting the same quiz twice. Not tested.

TC#	What to Test	Input	Expected Output
UT-NEW-11	Quiz status — not yet attempted	Valid student token + topic name	{ "attempted": false }
UT-NEW-12	Quiz status — already submitted	Same student + same topic (after submission)	{ "attempted": true }
IT-NEW-01	Submit quiz twice (integration)	Score POST → check status → POST again	Second submit blocked or flagged by system

B.4 — Topic History & Due-Time Expiry (timetable.py Lines 113–132, 321–347)

TC#	What to Test	Input	Expected Output
UT-NEW-13	Get topic history for year 1	GET /timetable/history?year=1	Returns list of all past topics, newest first
UT-NEW-14	Set topic twice — both rows logged	Two set-today calls for year 2	History has 2 rows; DailyTopic has 1 (upserted)
ST-NEW-01	Topic marked expired after due time	set-today with due_time = past time	"is_expired": true in get-today response
ST-NEW-02	Topic not expired before due time	set-today with due_time = 2 hours ahead	"is_expired": false in get-today response

B.5 — Mentor Registration (auth.py Lines 144–165)

TC#	What to Test	Input	Expected Output
UT-NEW-17	Mentor register — valid data	POST /auth/mentor/register with name, dept, identity, password	201 Created — new mentor record
UT-NEW-18	Mentor register — duplicate identity	Same identity (email/ID) used again	400 Mentor identity already registered

B.6 — Weekly Timetable Save by Mentor (timetable.py Lines 265–296)

TC#	What to Test	Input	Expected Output
IT-NEW-02	Mentor saves full weekly timetable	POST /timetable/weekly with year=2 + full schedule JSON	Old records deleted, new records saved correctly
IT-NEW-03	Student views updated timetable	Student logs in, opens timetable after mentor update	Shows newly saved subjects and topics

B.7 — Gemini AI Fallback Behavior (chat.py Lines 74–126)

TC#	What to Test	Condition	Expected Output
UT-NEW-20	Chatbot responds when Gemini is down	Invalid / missing GEMINI_API_KEY in .env	Fallback reply returned, no server crash
UT-NEW-21	Off-topic messages are redirected	Student asks about unrelated topic	AI replies: 'I can only help with [assigned topic] today'

B.8 — Other Missing Tests

TC#	What to Test	Where	Expected Output
UT-NEW-15	Get learning resources for topic	GET /students/resources + valid token	Returns resource list for assigned topic
UT-NEW-16	Resources endpoint — unauthenticated	No auth token	401 Unauthorized
UT-NEW-19	Welcome email triggered on register	POST /auth/student/register (new account)	Email queued to Resend/SMTP for new student
ST-NEW-03	CORS wildcard check	Review allow_origins in main.py	Should be restricted to frontend domain in production

C. Suggestions to Improve Existing Tests

C.1 — Add FAIL Cases to the Document

Every existing test currently shows PASS. A complete review document should include 2–3 intentional failures to demonstrate the test suite actually catches bugs.

TC#	Input	Expected	Actual (Before Fix)	Result
UT-15	score = -5	422 Error	Score saved incorrectly	FAIL → Fixed
IT-05	No token on mentor route	403 Forbidden	200 OK (no auth check)	FAIL → Fixed

C.2 — Equivalence Partitioning Table Has Gaps

Field	Current EP Coverage	Missing / Gap
Year	1, 2, 3 (string)	Integer 1 sent instead of string — what happens?
Topic Name	Any text, 1 char, 255 chars	Unicode / emoji (topics use emoji: 'Variables ■')
Password	Less than 8 chars	Password = only spaces (should fail but may not)
Email	Valid / invalid format	Case sensitivity: ARUN@EMAIL.COM vs arun@email.com

C.3 — Integration Tests Have No Failure Paths

All 6 integration tests (IT-01 to IT-06) cover only the happy path. Add at least one failure-scenario integration test.

TC#	Scenario	Modules Connected	Expected
IT-07	Student tries to access mentor dashboard	auth → students → MentorDashboard	403 Forbidden
IT-08	Mentor sets topic for invalid year (e.g., year=5)	timetable → DailyTopic	Graceful fallback or validation error

C.4 — Add Non-Functional Requirements (NFR) to Validation Section

NFR#	Requirement	How to Test
NFR-01	All passwords must be hashed — never stored in plaintext	DB inspection: password_hash starts with $2b$ (bcrypt)
NFR-02	All API responses within 500 ms under normal load	Timer on each endpoint using Postman / pytest
NFR-03	CORS headers correctly restrict origins in production	cURL or browser devtools check for allowed origins
NFR-04	JWT tokens expire after configured duration	Wait past token expiry, retry request — expect 401

C.5 — Expand Security Testing Section

■ IMPORTANT: The in-memory OTP store (_otp_store in auth.py) resets on every server restart. Any pending OTP is lost. This architectural risk must be documented.

TC#	What to Test	How Tested	Expected
ST-NEW-04	Brute-force login attempts	Send 50 wrong password requests to /auth/student/login	No rate limiting — document as known risk
ST-NEW-05	OTP cleared on server restart	Request OTP → restart server → use OTP	400 No OTP found — OTP lost (document risk)
ST-NEW-06	CORS wildcard in production	Check main.py allow_origins=["*"]	Should be restricted to frontend domain only
ST-NEW-07	Mentor token registers another mentor	Use mentor JWT to POST /auth/mentor/register	Should require admin auth or be restricted

C.6 — Increase Concurrent Users in Performance Testing

Metric	Current Target	Current Actual	Recommended Target
Concurrent users	10+	15 tested	25+ (realistic class: 21 students seeded)
AI chat under load	Not tested	Not tested	5 simultaneous Gemini requests
DB with 1000 score records	Not tested	Not tested	Query time < 300 ms
Timetable weekly fetch	Not tested	Not tested	< 200 ms for full 3-year schedule

D. Summary of All Missing Tests

Category	Missing Tests	Priority
OTP / Forgot Password (auth.py)	7 tests	HIGH
Quiz Generation & Status (students.py)	3 tests	HIGH
Security Gaps (CORS, Brute-force, OTP memory)	4 tests	HIGH
Topic History & Due-Time Expiry (timetable.py)	4 tests	MEDIUM
Mentor Registration (auth.py)	2 tests	MEDIUM
Weekly Timetable Save (timetable.py)	2 tests	MEDIUM
Gemini AI Fallback (chat.py)	2 tests	MEDIUM
Learning Resources Endpoint	2 tests	LOW
Welcome Email on Registration	1 test	LOW
TOTAL	27 tests	

E. Recommended Updated Test Count for Review 3

Testing Level	Current TCs	New to Add	Recommended Total	Pass Target
Unit Testing	16	+21	37	37
Integration Testing	6	+3	9	9
Validation Testing	13	+4	17	17
System Testing	11	+7	18	18
TOTAL	46	+35	81	81

■ QUICK WIN for Review 3: Add the 7 OTP tests + 2 quiz-status tests + 3 security gap tests = 12 high-priority tests in one session. These cover real working endpoints that currently have zero documentation.

Reviewer Note: The existing document is well-written and demonstrates good testing discipline. The 27 missing tests are not failures of the current work — they are features that were implemented but not yet documented in the test plan. Prioritise the OTP, quiz-status, and security gap tests before Review 3 as these cover production-ready endpoints and known architectural risks.